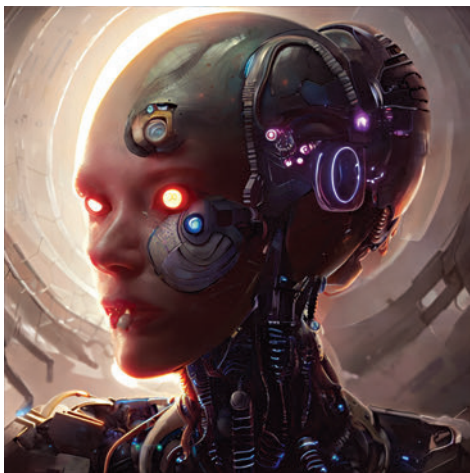


images by adding Gaussian noise to an original image, and then training a model to predict the noise that was added to the image from one step to the next. As the steps progress, the image becomes more and more degraded by the added noise until the last step where it is completely degraded. However, if the steps are small enough, the final image can be approximated by the same Gaussian distribution as the added noise. This allows us to generate new images by sampling from that same distribution and passing it through the trained model N times.

Guiding the Diffusion Process

We've seen how diffusion models can generate images from random noise, but this alone would only produce random images. To generate images corresponding to specific class labels, text embeddings, or images, we must build upon two key concepts proposed in earlier works:



First, conditioned diffusion involves feeding conditioning data such as class labels or text embeddings into the diffusion model's input and inner layers through attention to aid in producing samples that match the input data.

Second, classifier guidance involves adding the gradient with respect to the predicted target class from a classifier to the noise generated by the diffusion model, thereby steering the diffusion process towards the desired class at each step.

There are two ways to do this.

CLIP guidance is similar to classifier guidance, but instead of adding the gradient with respect to a target class predicted by a classifier, it sums the gradient with respect to the CLIP loss between the partially-generated image and the caption. The CLIP loss measures the correspondence between the image and the caption. If you're not familiar with CLIP, we recommend checking it out on OpenAI's blog.